

OPTIMUS PRIME / EXPEDITION SKILLS FOR HUMANOID MOUNTAINEERING

THE MOUNTAIN AMPLIFIES FAILURE

A SLIP BECOMES A FALL.

A FALL BECOMES A RESCUE.

G1 must use tools, recover, and know when to stop.



SIMPLIFIED

OPTIMUS PRIME / EXPEDITION SKILLS FOR HUMANOID MOUNTAINEERING

A SLIP BECOMES A RESCUE.

G1 MUST RECOVER BEFORE A HUMAN MUST INTERVENE.

Extreme terrain turns small failures into high-consequence missions.



FOUR SKILLS FOR EXTREME TERRAIN.

ONE EXPEDITION SYSTEM / CURRICULUM-TRAINED POLICIES



01 SELF-ARREST

STOP THE
SLIDE



02 FALL RECOVERY

GET BACK
UP



03 CONTROLLED RAPPEL

DESCEND WITH
CONTROL



04 TEAM LIFT

MOVE HEAVIER
LOADS TOGETHER

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HOW SINGLE-ROBOT SKILLS LEARN

MUJOCO / PPO / HIERARCHICAL CONTROL

SELF-ARREST PPO

125-D OBSERVATION →

2 × 256 TANH →

14 ARM RESIDUALS

100 Hz · ≈200k actor+critic params

FIXED-LINE RECOVERY

FALL + ROPE STATE →

2 × 256 TANH →

4-ACTION HANDOFF

50 Hz · pretrained 29-DoF WBC prior

REWARD SHAPING / PHYSICS MUST EXPLAIN SUCCESS

01 TOOL CONTACT

rigid pick contact
blade 22–42° · snow load

02 BODY CONTROL

chest-down pose · grip
wrists stay ahead of torso

03 TERMINAL GATES

no passive stop
stable stand earns success

CURRICULUM

0.25–0.75 m/s

START EASY

→ 5.0 m/s

SPEED

→ ±40° / ±1.5 m/s

HEADING + CROSS-SLOPE

→ HARD ANCHORS

ROLL + ADVERSARIAL FALLS

→ 9/9 + 60/60

FROZEN SELECTION

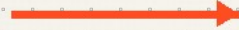
SIMPLIFIED

ONE POLICY. MANY FALLS.

MUJOCO SIMULATION / PPO / REWARD SHAPING

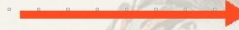
125 OBSERVATIONS

pose · speed · tool load



MUJOCO + PPO

100 Hz compact policy



14 ARM COMMANDS

plant and load the axe

PHYSICAL REWARD SHAPING IN MUJOCO

REWARDS REQUIRE RIGID PICK CONTACT, 22–42° BLADE ANGLE, AND REAL SNOW LOAD

CURRICULUM LEARNING PROGRESSION

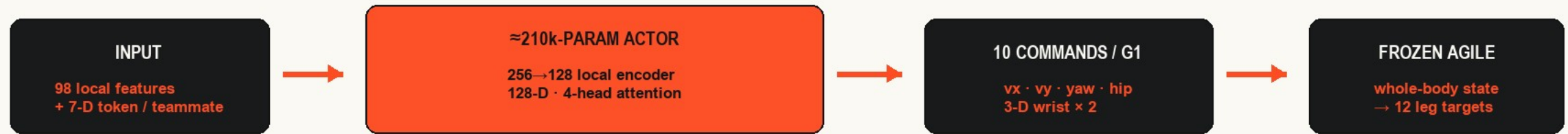
EASY SLIDES → 5 m/s + CROSS-SLOPE + ADVERSARIAL FALLS

SUCCESS COUNTS ONLY WHEN THE AXE PHYSICALLY BITES.

HOW THE TEAM POLICY LEARNS

ISAAC SIM / ISAAC LAB / SHARED MAPPO / CURRICULUM

SHARED ATTENTION ACTOR / SAME WEIGHTS ON EVERY G1



TRAINING-ONLY CENTRAL CRITIC

TEAM STATE + PAYLOAD MASS

768 → 512 → 256

ACTOR: LOCAL + TOKENS ONLY
AT INFERENCE

MAPPO: 24-step rollouts · 5 epochs · $\gamma .99$ · GAE .95 · clip .2 · lr 3e-4

REWARD SHAPING / SHARED TEAM OBJECTIVE

+ TRACK

+ LEVEL

+ HEADING

+ LOAD BALANCE

+ UPRIGHT

+ LIFT PROGRESS

- EXTENSION / RATE / FALL

CURRICULUM

LIFT + LEVEL

0-24k steps

ADD CARRY + TURN

24k→150k

MASS 8→18 kg

by 180k

2 / 3 / 5 G1

shared actor

ONE SHARED POLICY. EVERY ROBOT.

ISAAC SIM & ISAAC LAB / COOPERATIVE MAPPO



LOCAL STATE + TEAM TOKENS

98 local features + 7-D token / teammate



ISAAC SIM GPU ACTOR (MAPPO)

10 COMMANDS / G1 · FROZEN AGILE LEGS

TEAM REWARD SHAPING:

+ TRACK + LEVEL + LOAD BALANCE - RATE / FALLS

Same actor weights on every robot.

MULTI-AGENT CURRICULUM IN ISAAC SIM

LIFT FIRST → CARRY + TURN → HEAVIER LOADS (8→18 kg)

GPU-VECTORIZED SIMULATION TRAINS DECENTRALIZED COORDINATION

LIVEKIT CONNECTS THE EXPEDITION

ONE ROOM / TWO EXPLICIT CONTROL PLANES

ROBOT STATE PLANE / LIVEKIT DATA TRACKS



OPERATOR VOICE PLANE / LIVEKIT AGENTS + INFERENCE



VOICE SUPERVISES · LOCAL POLICIES CONTROL MOTION

LIVEKIT IS THE EXPEDITION BUS.

STATE + VOICE

STATE

ONE STATE STREAM / G1



LIVEKIT ROOM



EVERY ACTOR STAYS IN SYNC

VOICE

"MOVE THE LOG"



CONFIRM



GATED START INTENT

VOICE SUPERVISES. LOCAL POLICIES CONTROL MOTION.

WHAT WE MEASURED

FIXED TESTS / PHYSICAL OUTCOMES



SELF-ARREST

9 / 9 SCENARIOS

60 / 60 unseen falls arrested



FALL RECOVERY

RECOVERS TO STAND

then continues uphill



RAPPEL

2.00 m DESCENT

7 / 10 randomized starts passed



TEAM LIFT

50 / 50 SHARE

overload or imbalance → abort

FOUR SKILLS. MEASURED RESULTS.

FROZEN EVALUATION

ARRESTS PASSED

69 / 69

9 named + 60 unseen

ICY FIXED LINE

STANDS AGAIN

recover + continue

RAPPEL DESCENT

2.00 m

controlled braking

TARGET LOAD SPLIT

50% / 50%

balanced teamwork

STOP · RECOVER · DESCEND · COORDINATE

SIM-TO-REAL: OPEN CHALLENGES

SCALING TO THE HIMALAYAS / THE C.L.I.M.B. FRONTIER

C**CONTACT**

Snow & Ice Dynamics

- Deformable snow crust
- Non-linear friction drops
- Slush & sliding scree
- Dynamic soft terrain

L**LIGHTING**

Optics & Sensors

- Flat-light whiteout
- Extreme snow glare
- Harsh cliff shadows
- Precipitation LiDAR clutter

I**INTER-COMMS**

Field Mesh Network

- No cloud LiveKit at altitude
- Ad-hoc local wireless mesh
- Packet drops >150 ms deadline
- Relative UWB sensing

M**MECHANICAL**

Ropes & Tethers

- Dynamic rope elasticity
- Variable carabiner friction
- Dynamic line bounce & sag
- Line snag & twist hazards

B**BALANCE**

Unassisted Authority

- Withdraw sim balance assists
- Whole-body 43-DoF torque
- Sub-zero battery drop
- Thermal actuator limits

BRIDGING THE GAP FROM SIMULATION TO HARSH ALPINE DEPLOYMENT

SIM-TO-REAL: OPEN CHALLENGES

THE C.L.I.M.B. ROADMAP FOR ALPINE ROBOTICS

C

CONTACT

SNOW & ICE

Slush, crust
fracture, and non-
linear friction
drops on steep
slopes

L

LIGHTING

OPTICS & SENSORS

Flat-light
whiteout
blindness, intense
snow glare, and
LiDAR backscatter

I

INTER-COMMS

OFFLINE MESH

Ad-hoc peer-to-
peer radio mesh
when cloud
connectivity is
unavailable

M

MECHANICAL

ROPES & TETHERS

Rope elasticity,
carabiner friction
variation, and
dynamic line
bounce

B

BALANCE

UNASSISTED TORQUE

Replacing sim-only
stabilizer
crutches with true
whole-body motor
control

CONTACT · LIGHTING · INTER-COMMS · MECHANICAL · BALANCE